

Q1 - 24 January - Shift 1

Given below are two statements :

Statement I : If the Brewster's angle for the light propagating from air to glass is θ_B , then Brewster's angle for the light propagating from glass to air is $\frac{\pi}{2} - \theta_B$.

Statement II : The Brewster's angle for the light propagating from glass to air is $\tan^{-1}(\mu_g)$ where μ_g is the refractive index of glass.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both Statements I and Statement II are true.
- (2) Statement I is true but Statement II is false.
- (3) Both Statement I and Statement II are false.
- (4) Statement I is false but Statement II is true.

Space for your notes:

Q2 - 25 January - Shift 1

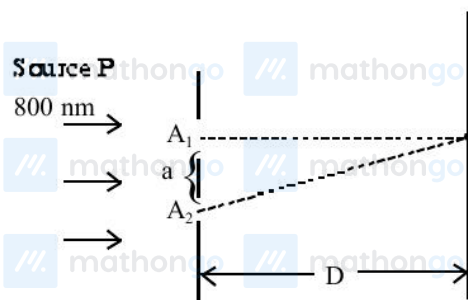
In Young's double slits experiment, the position of 5th bright fringe from the central maximum is 5 cm. The distance between slits and screen is 1 m and wavelength of used monochromatic light is 600 nm. The separation between the slits is:

- | | |
|----------------------|----------------------|
| (1) 60 μm | (2) 48 μm |
| (3) 12 μm | (4) 36 μm |

Space for your notes:

Q3 - 29 January - Shift 1

In a Young's double slit experiment, two slits are illuminated with a light of wavelength 800 nm. The line joining A_1P is perpendicular to A_1A_2 as shown in the figure. If the first minimum is detected at P, the value of slits separation 'a' will be :

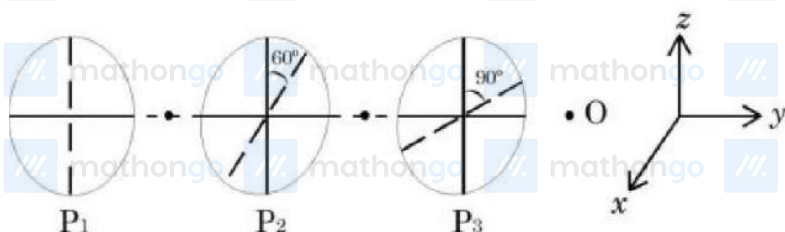


The distance of screen from slits $D = 5 \text{ cm}$

- (1) 0.4 mm (2) 0.5 mm
(3) 0.2 mm (4) 0.1 mm

Q4 - 29 January - Shift 1

As shown in figures, three identical polaroids P_1 , P_2 and P_3 are placed one after another. The pass axis of P_2 and P_3 are inclined at angle of 60° and 90° with respect to axis of P_1 . The source S has an intensity of $256 \frac{W}{m^2}$. The intensity of light at point O is $\frac{W}{m^2}$.



Q5 - 29 January - Shift 2

Unpolarised light is incident on the boundary between two dielectric media, whose dielectric constants are 2.8 (medium –1) and 6.8 (medium – 2), respectively. To satisfy the condition, so that the reflected and refracted rays are perpendicular to each other, the angle of incidence should be

$\tan^{-1}\left(1 + \frac{10}{\theta}\right)^{\frac{1}{2}}$ the value of θ is _____.

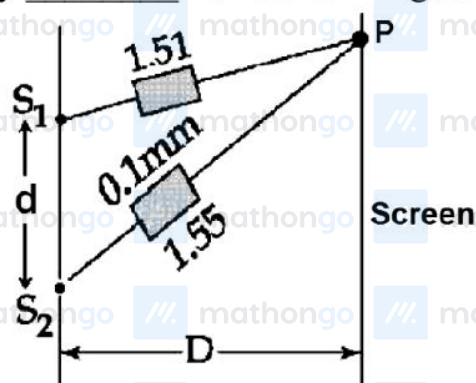
(Given for dielectric media, $\mu_r = 1$)

Space for your notes:

Q6 - 30 January - Shift 1

In Young's double slit experiment, two slits S_1 and S_2 are 'd' distance apart and the separation from slits to screen is D (as shown in figure). Now if two transparent slabs of equal thickness 0.1 mm but refractive index 1.51 and 1.55 are introduced in the path of beam ($\lambda = 4000\text{\AA}$) from S_1 and S_2 respectively. The central bright fringe spot will shift by _____ number of fringes.

Space for your notes:



Q7 - 30 January - Shift 2

In a Young's double slit experiment, the intensities at two points, for the path difference $\frac{\lambda}{4}$ and $\frac{\lambda}{3}$ (λ being the wavelength of light used) are I_1 and I_2 respectively. If I_0 denotes the intensity produced by each one of the individual slits, then $\frac{I_1 + I_2}{I_0} = \dots\dots$

Space for your notes:

Q8 - 31 January - Shift 1

Two polaroids A and B are placed in such a way that the pass-axis of polaroids are perpendicular to each other. Now, another polaroid C is placed between A and B bisecting angle between them. If intensity of unpolarised light is I_0 then intensity of transmitted light after passing through polaroid B will be:

Space for your notes:

- (1) $\frac{I_0}{4}$ (2) $\frac{I_0}{2}$
(3) $\frac{I_0}{8}$ (4) Zero

Q9 - 31 January - Shift 1

Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: The beam of electrons shows wave nature and exhibit interference and diffraction.

Reason R : Davisson Germer Experimentally verified the wave nature of electrons.

In the light of the above statements. Choose the **most appropriate** answer from the options given below :

- (1) A is correct but R is not correct
- (2) A is not correct but R is correct
- (3) Both A and R are correct but R is Not the correct explanation of A
- (4) Both A and R are correct and R is the correct explanation of A

Space for your notes:

Q10 - 31 January - Shift 2

Two light waves of wavelengths 800 and 600 nm are used in Young's double slit experiment to obtain interference fringes on a screen placed 7 m away from plane of slits. If the two slits are separated by 0.35 mm, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be mm.

Space for your notes:

Q11 - 01 February - Shift 1

' n ' polarizing sheets are arranged such that each makes an angle 45° with the proceeding sheet. An unpolarized light of intensity I is incident into this arrangement. The output intensity is found to be $\frac{I}{64}$. The value of n will be:

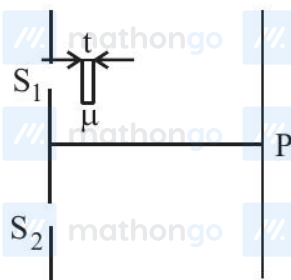
- (1) 3 (2) 6
(3) 5 (4) 4

Space for your notes:

Q12 - 01 February - Shift 2

As shown in the figure, in Young's double slit experiment, a thin plate of thickness $t = 10 \mu\text{m}$ and refractive index $\mu = 1.2$ is inserted in front of slit S_1 . The experiment is conducted in air ($\mu = 1$) and uses a monochromatic light of wavelength $\lambda = 500 \text{ nm}$. Due to the insertion of the plate, central maxima is shifted by a distance of $x\beta_0$. β_0 is the fringe-width before the insertion of the plate. The value of the x is_____.

Space for your notes:



Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2)

Q2 (1)

Q3 (3)

Q4 (24)

Q5 (7)

Q6 (10)

Q7 (3)

Q8 (3)

Q9 (4)

Q10 (48)

Q11 (2)

Q12 (4)

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Q1 (2)



$$\mu_a \sin i_1 = \mu_g \sin(90 - i_1)$$

$$\tan i_1 = \frac{\mu_g}{\mu_a}$$

When going from glass to air

$$\tan i_2 = \frac{\mu_a}{\mu_g} = \cot i_1$$

Hence

$$i_2 = \frac{\pi}{2} - i_1$$

Q2 (1)

Given

$$D = 1\text{m}$$

$$\lambda = 600 \times 10^{-9}\text{m}$$

$$n = 5$$

$$\text{As } y_{\text{nth}} = \frac{n\lambda D}{d}$$

$$\Rightarrow \frac{5 \times 600 \times 10^{-9} \times 1}{d} = 5 \times 10^{-2}$$

$$\Rightarrow d = \frac{5 \times 600 \times 10^{-9} \times 1}{5 \times 10^{-2}} = 60 \times 10^{-6}\text{m}$$

$$\Rightarrow d = 60\text{ }\mu\text{m}$$

Q3 (3)

$$A_2P - A_1P = \frac{\lambda}{2} \quad (\text{Condition of minima})$$

$$\sqrt{D^2 + a^2} - D = \frac{\lambda}{2}$$

$$D \left(1 + \frac{a^2}{D^2} \right)^{1/2} - D = \frac{\lambda}{2}$$

$$D \left(1 + \frac{1}{2} \times \frac{a^2}{D^2} \right) - D = \frac{\lambda}{2}$$

$$\frac{a^2}{2D} = \frac{\lambda}{2} \Rightarrow a = \sqrt{\lambda \cdot D}$$

$$= \sqrt{800 \times 10^{-6} \times 50}$$

$$a = 0.2\text{ mm}$$

Q4 (24)

By first polaroid P1 intensity will be halved then P2 and P3 will make intensity $\cos^2(60^\circ)$ and $\cos^2(30^\circ)$ times respectively.

$$\text{Intensity out} = \frac{256}{2} \times \frac{1}{4} \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{256 \times 3}{2 \times 4 \times 4} = 24$$

Q5 (7)

$$\mu_1 = \sqrt{2.8 \times 1} = \sqrt{2.8}$$

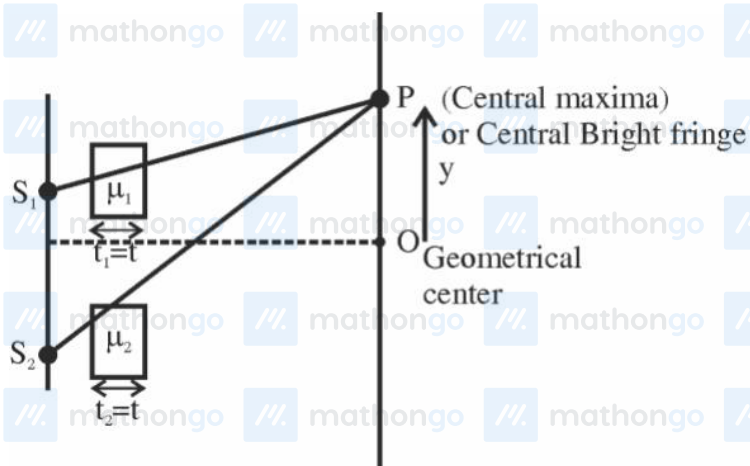
$$\mu_2 = \sqrt{6.8 \times 1} = \sqrt{6.8}$$

$$\mu_1 \sin i = \mu_2 \cos i \quad \tan i = \frac{\mu_2}{\mu_1} = \sqrt{\frac{6.8}{2.8}}$$

$$\tan i = \left(\frac{2.8 + 4}{2.8}\right)^{1/2} \quad i = \tan^{-1}\left(1 + \frac{10}{7}\right)^{1/2}$$

$$\theta = 7 \text{ Ans.}$$

Q6 (10)



Path difference at P be Δx

$$\Delta x = (\mu_2 - \mu_1)t$$

$$= (1.55 - 1.51)0.1 \text{ mm}$$

$$= 0.04 \times 10^{-4}$$

$$\Delta x = 4 \times 10^{-6} = 4 \mu\text{m}$$

$$y = \frac{\Delta x D}{d} = 4 \times 10^{-6} \frac{D}{d}$$

{y is the distance of central maxima from geometric center}

$$\text{fringe width } (\beta) = \frac{\lambda D}{d} = 4 \times 10^{-6} \text{ m } \frac{D}{d} = 4 \mu\text{m } \frac{D}{d}$$

$\therefore$ Central bright fringe spot will shift by 'x'

$$\text{Number of shift} = \frac{y}{\beta}$$

$$= \frac{4 \times 10^{-6} D / d}{4 \times 10^{-7} D / d} = 10 \text{ Ans}$$

Q7 (3)

$$I = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$I_1 = 4I_0 \cos^2\left(\frac{\pi}{4}\right) = 2I_0$$

$$I_2 = 4I_0 \cos^2\left(\frac{2\pi}{3}\right) = I_0$$

$$\Rightarrow \frac{I_1 + I_2}{I_0} = 3$$

Q8 (3)

$$I_A = \frac{I_0}{2}$$

$$I_C = \frac{I_0}{2} \cos^2 45 = \frac{I_0}{4}$$

$$I_B = I_C \cos^2 45 = \frac{I_0}{8}$$

Option 3.

Q9 (4)

Conceptual

Q10 (48)

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No Solution

Q11 (2)

After passing through first sheet

$$I_1 = \frac{I}{2}$$

After passing through second sheet

$$I_2 = I_1 \cos^2(45^\circ) = \frac{I}{4}$$

After passing through n^{th} sheet

$$I_n = \frac{I}{2^n} = \frac{I}{64}$$

$$n = 6$$

Q12 (4)

Fringe shift = $\frac{\Delta x}{\lambda} B$

$$= \frac{10 \times 10^{-6} (1.2 - 1)}{5 \times 10^{-7}} B$$

$$= \frac{10^{-5} \times 0.2}{5 \times 10^{-7}} = 4$$